ReCycleGuard

An Agentic Assistant for Auditable E-Waste EPR Compliance Reconciliation

1BM23CD020 H Sampreeth Bhat 1BM23CD047 R V Abhishek (Team Lead)
1BM23CD063 Sushanth S 1BM23CD069 Vedantha Vinay Vernekar

B.M.S. College of Engineering, Bengaluru
Department of CSE (Data Science)
Agentic AI (26CD7PEAAI), Final Review

November 2026

Problem and the decision supported

- Indian producers of electronics must meet recycling obligations under the E-Waste (Management) Rules, 2022 by buying EPR credits from registered recyclers.
- The officer must show that, for each metal, the obligation, certificates, transfers, payments and portal balance agree.
- The records live in PDFs, ledgers and a portal export, and they disagree in specific ways.

Decision supported

Which discrepancy should the compliance officer investigate next, and what verification or corrective action does the evidence support?

The output is an advisory working paper for a human reviewer. It is not a filing, a legal verdict or a fraud finding.

Why a record discrepancy is hard to read

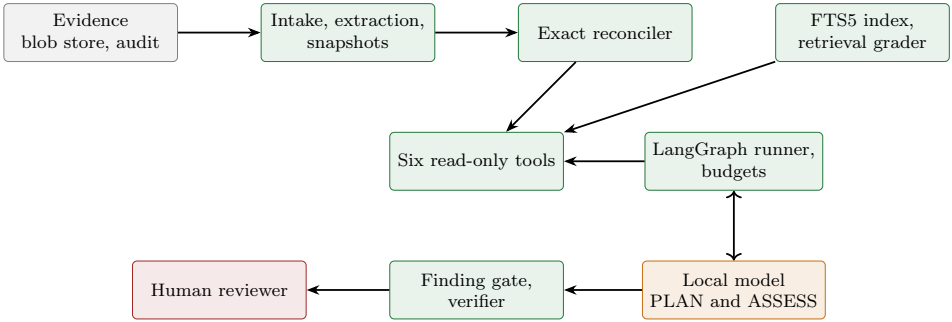
Typical faults

- overlapping exports count a transfer twice
- tonnes mixed into kilograms
- an unregistered entity alias on an invoice
- cancelled or expired certificates still counted
- payment made but the portal transfer never completed
- a memo read as an official deadline extension

Why not a script or one prompt

- A script cannot choose what to check next.
- A single prompt cannot calculate exactly or cite reliably.
- The useful number of steps varies by case, and a stopping rule is needed.

Architecture



Green: deterministic code. Orange: the model. Red: the human decision. The model proposes, code decides what is accepted.

The agent loop

1. **Preflight** (no model): missing pages stop the run.
2. **PLAN**: the model proposes one check or a stop.
3. **Act**: a whitelisted tool runs (exact decimals or retrieval).
4. **ASSESS**: the model judges the observation: supports, refutes, weak, no progress, unknown.
5. Continue, resolve or abstain. `RESOLVE` requires `SUPPORTS`.
6. Draft, verify (one repair), then **human review**.

Bounds

12 model calls, 600 s, token reservation checked before dispatch; 8 tool calls and 2 reformulations per finding; repeat actions stopped by a fingerprint.

Each turn is committed with a compare-and-swap version, so a killed run resumes without repeating a tool call.

Division of labour between the model and the code

	Model may	**Model may not**
Quantities	read results of exact tools	compute, state or alter a number
Evidence	choose what to inspect next	approve evidence or edit originals
Findings	attach one proposal	bypass the gate (citations, digits, wording)
Tools	pick from six read-only tools	run code, SQL, shell, or write
Decision	assess and abstain	act as reviewer or issue a legal verdict

Six audited tools

Tool	Purpose
`inspect_evidence`	parsed fields and exact locators of observations
`lookup_locator`	verified provision text at a legal path
`trace_certificate`	lifecycle: issued, transferred, cancelled, expired
`compute_reconciliation`	required, supported, gap, pending per bucket
`compare_return`	portal claims against ledger-supported balances
`search_rules`	BM25 passages graded for authority and date

Each call stores an intent record before it runs and a result record after it completes.

Safety and guardrails

- **Untrusted text:** dossier and retrieved text enter the prompt as escaped JSON; a benchmark family inserts an override instruction into an invoice remark.
- **Finding gate:** citations must be evidence this run returned; no digits, prohibited or authority wording.
- **No blind retry:** an uncertain model-call outcome ends the run; every attempt is kept.
- **Host sandbox:** macOS profile, loopback-only network, probes must all be denied before start (1,630 of 1,630 denied in the final run).
- **Human review:** the decision is bound to the exact report and input digests.

Data and evaluation protocol

Synthetic benchmark `pilot-0.1`

- 48 fictional dossiers, 16 fault families
- 24 development, 24 held-out (sealed, outside the repository)
- 8 combinations per split
- references are agent-validated, not hand-reviewed

Variants

- V0 deterministic, no model
- SC scripted control
- V1 fixed retrieval, real model
- V2 adaptive, real model

Primary metric: supported case success. Human rubric: citation, dates, next action (blind to variant).

Development reliability, prompt v9 (24 dossiers × V1, V2)

Criterion (threshold)	**V1**	**V2**
Runs not failed ($\geq 90\%$)	24/24	24/24
Schema-valid replies per call ($\geq 95\%$)	132/132	43/43
Call timeouts ($\leq 5\%$)	0	0
Forbidden actions executed (0)	0	0
Mean wall time (V1 $\leq$ 120 s, V2 $\leq$ 240 s)	134 s	62 s
Mean model calls per run	5.5	1.8
Correct model finding (≥ 1)	not met	

Development set only; a check of the setup, not a claim of quality. Crash tests: killed runner and killed server both recovered with no repeated trial.

Quality on development dossiers

	V1	V2
Exact reconciliation (reviewed trials)	28/28 buckets	26/26 buckets
Primary success (all 24 / reviewed only)	2 / 1 of 22	2 / 1 of 22
Micro precision / recall	0.67 / 0.21	0.67 / 0.15
Model proposals made / accepted / correct	16 / 1 / 0	0 / 0 / 0

- Reconciliation is exact because it is deterministic.
- Fault families found came from deterministic rule and scope checks (F11, F12, F14).
- 14 of 16 proposals were refused for authority wording; the one accepted finding was a false positive.

Prompt development on the tuning set

Version	Effect
v5, v6	invented ids: 29, then 40 of 48 not failed
v7	100% schema-valid, no proposals
v8	40 invalid attempts: proposal at the wrong level
v9	13 V1 proposals, 1 accepted

- Tuned only on 16 development cases; 8 checking cases run once afterwards.
- Each fix moved the failure to the next layer.
- Success rate did not change (2 of 16).

Experiment: fixed versus adaptive investigation

Hypothesis H1

On the same dossiers, tools, corpus and budget, V2 has a higher supported case-success rate than V1 at comparable cost.

- 24 held-out dossiers (two are expected preflight stops, reported with and without).
- Order of V1 and V2 alternates by case; repeats R fixed at the freeze, before any held-out run.
- Paired case-level bootstrap with all repeats of a case resampled together.
- Development evidence: V2 uses about one third of the model calls and is not more successful.

Held-out result

Variant	Success rate	95% interval	Calls per run
V0	pending	pending	–
V1	pending	pending	pending
V2	pending	pending	pending
V2 − V1		pending	

The held-out run waits for a frozen prompt and model setup. No held-out number is shown in this deck.

Limitations

- Synthetic data, a two-source corpus and one small quantised local model.
- Templates and seeds are in the repository, so the held-out set is not externally blind.
- One person built, ran and evaluated the system; blind teammate scoring reduces this, and does not remove it.
- The references were checked by a coding agent, not by a person.
- The model adds little on the development set; detections come mainly from deterministic checks.
- Multi-agent collaboration and memory across runs were not built.

Conclusion and next steps

- A bounded, audited agent loop with exact arithmetic, a gate for model output and mandatory human review.
- The setup is reliable on development data; quality of the model's findings is the open problem.
- **Next:** teach the prompt the substitutes for authority-list wording, tune on a faster runtime, freeze, and run the held-out experiment with the blind rubric.
- Final conclusion on adaptive investigation: **[held-out result: to be completed]**.

Team and ownership

USN	Name	Components
1BM23CD020	H Sampreeth Bhat	intake, extraction, index, retrieval, corpus
1BM23CD047	R V Abhishek (Lead)	orchestration, model gateway, sandbox, trial coordinator
1BM23CD063	Sushanth S	reconciler, tools, synthetic generator, dossiers
1BM23CD069	Vedantha Vinay Vernekar	audit kernel, reports, review, export, scoring isolation

Every member is expected to explain the whole system.

Questions

Repository: github.com/R-V-Abhishek/ReCycleGuard (branch `checkpoint-5`)